

GOOGLE PLAY STORE REVIEW ANALYTICS

Executive Summary

Metric	Value
Apps analyzed (post-cleaning)	8,892
User reviews scored for sentiment	37,427
Average app rating	4.19/5.0
Estimated cumulative installs	146.6 Billion
Estimated revenue (paid apps)	\$367.35 Million
Free apps	93.1%



Figure 1: Dashboard 1 (Executive Overview)

Figure 1 shows the core analytics dashboard produced from this analysis, summarizing catalogue composition, ratings, installs, and revenue at a glance.

The Google Play Store is one of the largest and most competitive digital marketplaces in the world, hosting millions of apps that compete daily for installs, attention, and revenue. For the people who manage that competition, publishers, product managers, and business analysts; the hard part isn't a shortage of data. It's that installs, ratings, reviews, and pricing all tell slightly different stories, and pulling them apart takes structured analysis rather than a glance at a Play Store listing page.

That's the gap this project sits in. Mobile app analytics sits right at the intersection of product strategy, marketing, and user experience and the metrics everyone already watches (installs, ratings, reviews) are only useful once someone has actually tested what they correlate with, and what they don't.

Key Takeaways

- This project turns raw Play Store data into a structured, decision-ready view of app performance.
- It replaces anecdotal impressions ("bigger apps get better reviews," "paid apps must perform better") with tested, data-backed answers.
- The output isn't just a notebook it's two interactive dashboards built for stakeholders who won't read the analysis line by line

Business Problem

Raw Play Store data, on its own, doesn't answer very much. App metadata arrives as inconsistently formatted text, and user reviews arrive as unstructured opinion, neither is directly usable for a business decision until it's been cleaned, structured, and interrogated. That gap is the actual problem this project solves.

Specifically, three questions sat unanswered before this analysis, and answering them precisely is the project's entire scope:

1. Reach vs. revenue: Which app categories actually lead on installs, and is that the same set of categories that leads on estimated revenue?
2. Sentiment vs. rating: Does what users *write* in their reviews agree with the star rating they *give*?
3. Rating predictability: Can an app's rating be explained, or predicted, from measurable attributes like installs, reviews, size, and price?

Business Impact

Without answers to these three questions, category investment, pricing strategy, and quality-improvement effort all end up guided by assumption rather than evidence. For example, treating the most-installed category as automatically the most valuable one, or treating "make the app smaller" as a default rating fix without checking whether size actually relates to rating at all.

Project Objectives

Prepare

- Clean and standardize raw app metadata and user review data into an analysis-ready form.
- Engineer derived features: log-scaled installs/reviews, rating tiers, estimated revenue, and update year to support meaningful comparison.

Analyze

- Explore the dataset through univariate, bivariate, and multivariate analysis to surface category-level and metric-level patterns.
- Quantify user review sentiment using VADER and test its relationship with star ratings.
- Build and honestly evaluate a predictive model for app rating.

Deliver

- Translate all findings into a business-insights narrative and an interactive dashboard.
- Implement six additional internship-assigned analytical tasks, each governed by its own business filters, multilingual labeling requirements, and time-restricted display logic as a second, dedicated dashboard.



Figure 2: Objective-to-Workflow Flow

Figure 2 shows how these objectives map onto the project's execution pipeline, from raw data through to the two delivered dashboards.

Key Takeaways

Objective Stage	What It Covers	Answered In
Prepare	Cleaning, feature engineering	Chapters 5–6
Analyze	EDA, statistics, sentiment, modelling	Chapters 6–9
Deliver	Dashboards, internship tasks	Chapter 10-11

Understanding the Dataset

This project draws on two source files that describe the Play Store ecosystem from different angles: one row per app, and one row per review.

	Play Store Data.csv	User Reviews.csv
Grain	One row per app	One row per review (many rows per app)
Rows	10,841	64,295
Columns	13	5
Key fields	App, Category, Rating, Reviews, Size, Installs, Type, Price, Content Rating, Genres, Last Updated, Current Ver, Android Ver	App, Translated_Review, Sentiment, Sentiment_Polarity, Sentiment_Subjectivity
Role in this project	Backbone of every EDA, statistical, and modelling chapter	Source for review-level sentiment scoring

Neither file arrived analysis-ready. A first-pass quality check surfaced five specific issues, each of which had to be resolved before anything downstream a chart, a correlation, a model could be trusted.

Issue	What It Looked Like	Why It Matters
Missing ratings	1,474 apps with no Rating value	Rating drives nearly every chart and the prediction model, unusable without it
Numbers stored as text	Installs ("10,000+"), Price ("\$4.99"), Reviews	Can't sort, sum, or chart a value pandas still reads as a string
Inconsistent size units	Mixed MB, KB, and "Varies with device"	Size comparisons across apps are meaningless without one common unit
Duplicate rows	483 fully duplicated app entries	Inflates every count and sum that follows
Missing review text	26,868 reviews with no text	Can't sentiment-score a review that has nothing written in it

Before vs. After Cleaning

	Play Store Data (apps)	User Reviews (scored)
Before	10,841	64,295
After	8,892	37,427

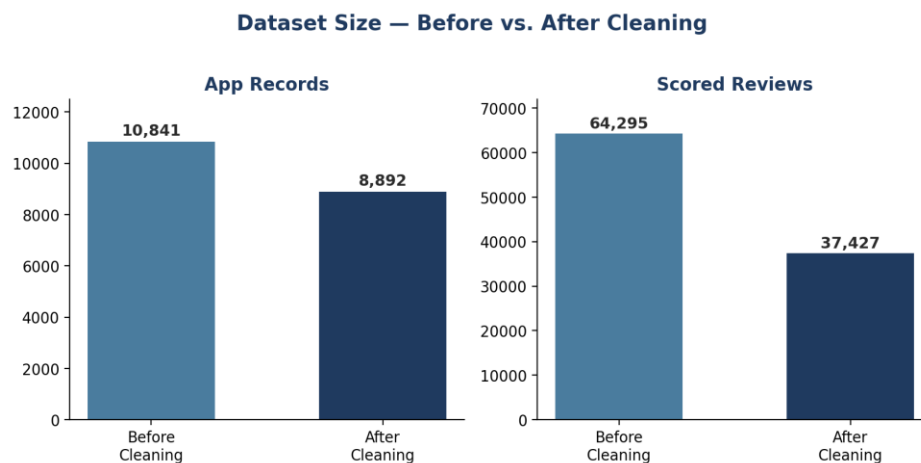


Figure 3: Dataset Size (Before vs. After Cleaning)

Figure 3 shows the scale of the cleaning effort, nearly 2,000 apps and 27,000 reviews were removed as unusable before analysis began.

Key Takeaways

- Two files, two different grains, app-level metadata and review-level text that had to be handled with separate cleaning logic before they could be merged.
- Five distinct, identifiable data-quality issues, not a vague "the data was messy."
- The working dataset used across every remaining chapter is 8,892 apps and 37,427 scored reviews a number worth remembering, since it appears again in almost every KPI throughout this report.

Preparing High-Quality Data

Missing Data

Fix	Reasoning
Dropped rows with missing Rating	Rating drives nearly every chart and the prediction model — not safely imputable
Filled remaining missing values with each column's mode	Applied only to non-critical categorical columns (Type, Content Rating, Current Ver, Android Ver), with affected columns explicitly logged rather than filled silently
Dropped reviews with no text	A review with nothing written in it can't be sentiment-scored

Type Conversion

Fix	Example
Installs: text → integer	"10,000+" → 10000
Price: text → float	"\$4.99" → 4.99
Reviews: text → integer	"3145" (string) → 3145 (int)

Unit Standardization (Size Conversion: Before → After)

Raw Value	Cleaned Value	Logic
"19M"	19.0	Strip "M", cast to float (already MB)
"512k"	0.5	Strip "k", divide by 1024 to convert KB → MB
"Varies with device"	<i>missing</i>	No real size known — left as missing rather than guessed

Duplicate & Invalid Removal

- 483 fully duplicated app rows dropped.
- 1 row with a Rating above the valid 0–5 range removed.

Post-Cleaning Validation

Rather than assuming these fixes worked, each one was re-checked with explicit assertions immediately afterward: zero missing values remaining in key columns, zero duplicate rows, correct numeric types on Installs/Price/Reviews, and zero invalid ratings left. If a future re-run on updated data ever broke one of these assumptions, it would fail loudly here not silently produce wrong numbers three chapters later.

Key Takeaways

- Every fix was tied to a specific, previously identified issue not applied generically.
- The trickiest transformation (Size) preserves genuine uncertainty ("Varies with device" → missing) instead of guessing a number.
- Every fix was independently re-verified with assertions before moving forward, this is what makes the 8,892-app dataset trustworthy for everything that follows.

Exploring the Data



Figure 4: Dashboard 1 (Catalogue Overview)

Figure 4 shows the catalogue's baseline composition — category concentration, the free-vs-paid split, and the overall rating distribution.

- **Category concentration:** a small number of categories dominate the catalogue; FAMILY leads by a wide margin.
- **Free vs. paid:** 93.1% of apps are free; paid apps are the exception, not the norm.
- **Rating distribution:** right-skewed, clustering between 4.0 and 4.5, with a median of 4.30, most published apps clear a basic quality bar.
- **Update activity:** a rising trend in the number of app updates over time.
- **Genre spread:** the top 10 genres by count, largely mirroring the category concentration above.

Key Takeaways

The catalogue skews heavily toward free apps and high ratings consistent with a mature, competitive marketplace where most published apps already clear a basic quality bar. This baseline matters for everything that follows: any category or app that stands out has to stand out *against* an already-crowded, already-well-rated field.

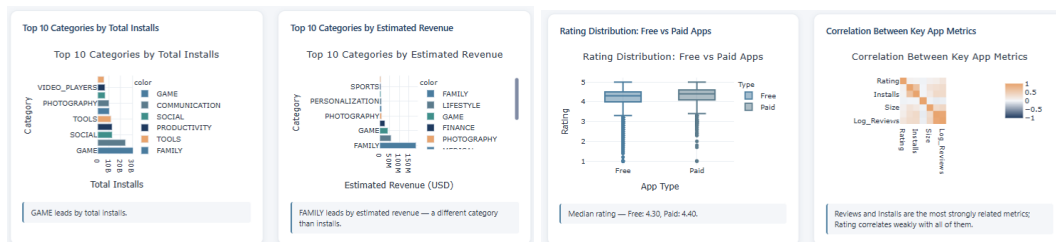


Figure 5: Dashboard 1 (Category Leaders & Rating Comparison)

Figure 5 compares category rankings by installs and by estimated revenue side by side, showing the divergence summarized in the table below.

Category Leaders — Installs vs. Revenue

Metric	Leading Category
Total Installs	GAME
Estimated Revenue	FAMILY

Business Meaning, Impact, Decision & Action

- **Business Meaning:** The category driving the most downloads is not the category generating the most estimated revenue.
- **Business Impact:** A category-investment decision made purely on install volume would misallocate spend. GAME's reach doesn't translate into FAMILY's revenue leadership.
- **Decision Supported:** Category strategy should be split into two separate tracks, one optimized for reach, one for monetization rather than one combined "top category" call.
- **Potential Action:** Budget owners evaluating category expansion should check both metrics independently before committing spend, rather than defaulting to whichever ranking is more visible.

Two smaller patterns rounded out this stage of the exploration: update recency showed no clear visual relationship with rating, and paid apps carried a modestly higher median rating (4.40) than free apps (4.30) a gap worth noting, but too small on its own to treat pricing as a reliable quality signal.

Statistical Validation

Visual patterns can be persuasive without being real two lines that look like they move together on a chart don't always hold up numerically. To check, every key numeric metric: Rating, Reviews, Installs, Price, Size, and their log-scaled versions was run through a Pearson correlation matrix, the standard measure of linear co-movement between continuous variables.

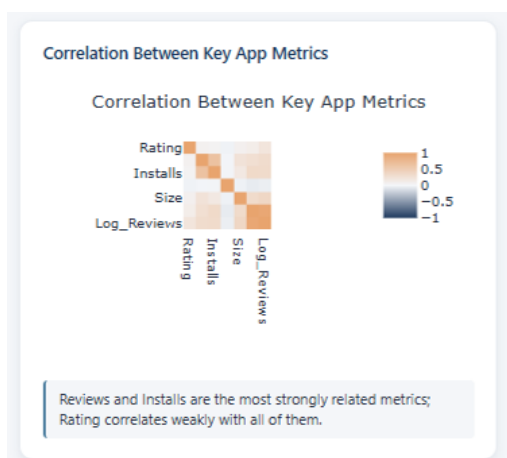


Figure 6: Correlation Heatmap (Key App Metrics)

Figure 6 shows the correlation heatmap across all key numeric metrics — the strong Reviews-Installs relationship and Rating's uniformly weak row are both visible at a glance.

- **Confirmed:** Reviews and Installs are, by a clear margin, the most strongly correlated pair in the matrix. Apps with more installs reliably accumulate more reviews. This is intuitive (more users, more chances to leave a review) and not, on its own, a business insight so much as a sanity check that the data behaves as expected.
- **Undercut:** Rating showed only a weak correlation with every other metric in the matrix not just installs, but reviews, price, and size too. None of the factors already explored on their own meaningfully explains why one app is rated higher than another.

Key Takeaways

- Installs and Reviews move together strongly: expected, and confirms the data is behaving sensibly.
- Rating is not well explained by any single metric already explored, a signal, not a dead end, for what's coming in the modelling chapter.

Sentiment Analysis

A user can leave three stars and still write an angry paragraph, or leave five stars with a lukewarm one-line comment. Star ratings and review tone are two different signals, and until they're measured separately, there's no way to know how closely they actually track each other.

To measure tone directly, every review was scored using VADER, a rule-based sentiment tool suited to short, informal text like app reviews. Before scoring all 37,427 reviews, VADER was run on three example sentences to confirm what its output actually looks like:

How VADER Scores Text

Example Review	VADER Read
"This app is amazing! I love the new features."	Clearly positive
"This app is very bad! I hate the new features."	Clearly negative
"This app is okay."	Neutral

Each review then received a single Sentiment Score: VADER's compound rating, running from -1 (very negative) to +1 (very positive).

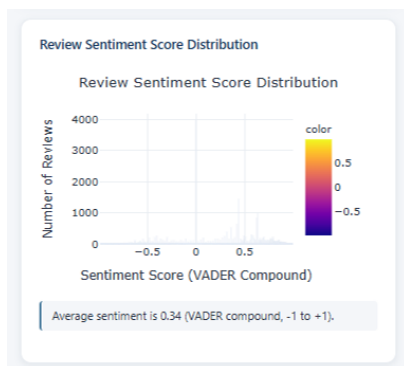


Figure 7: Review Sentiment Score Distribution

Figure 7 shows the distribution of sentiment scores across all 37,427 scored reviews, skewing moderately positive.

Across all scored reviews, the average sentiment score came out to 0.34, moderately positive on a scale where 0 is neutral and 1 is maximally positive. But averaged to the app level and compared against each app's actual star rating, the two signals turned out to track each other only loosely echoing the weak Rating correlations already surfaced before.

Business Meaning, Impact, Decision & Action

- **Business Meaning:** The tone of what users write and the star score they give are two different signals, not two versions of the same one.
- **Business Impact:** A category or app relying on star ratings alone could be sitting on written complaints — or written praise — that its star average doesn't reflect.
- **Decision Supported:** Sentiment and star rating should be monitored as independent metrics, not collapsed into one "user satisfaction" number.
- **Potential Action:** Product and support teams should treat a review-sentiment dip as an early warning worth investigating on its own, even if the star average hasn't moved yet.

Key Takeaways

- Reviews skew moderately positive on average (0.34 on a -1 to +1 scale).
- Sentiment and star rating are only weakly related confirming, from an entirely different angle, the same pattern Chapter 7's correlation matrix already showed for Rating.

Predictive Modelling

A Random Forest Regressor was trained on the features already available from feature engineering: Log_Installs, Log_Reviews, Price, Size, and a Free/Paid flag, using a standard 80/20 train-test split, so performance is reported only on data the model never saw during training.

R^2 measures how much of the variation in Rating a model explains. A score of 1.0 would mean the model explains everything; 0.0 means it explains nothing more than just guessing the average rating for every app; a *negative* score means it performs slightly *worse* than that simple guess. Model performance: $R^2 \approx -0.036$.

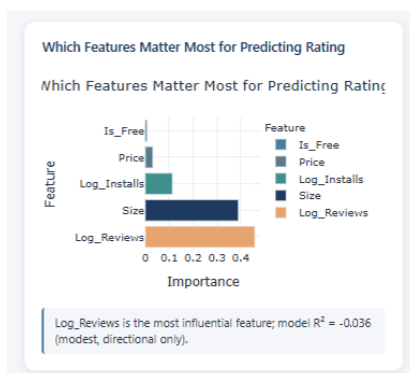


Figure 8: Feature Importance (Predicting Rating)

Figure 8 shows the relative feature importance from the model — Log_Reviews ranks highest, though no feature shows strong predictive value.

Log_Reviews came out as the most influential feature relative to the others in the model, but "most influential" here means least unhelpful, not genuinely predictive.

Key Findings

This result is not a modelling failure, it's a confirmation. Earlier we studied that Rating doesn't correlate strongly with structured metrics or with review sentiment. A near-zero R^2 is simply what that finding looks like when tested directly with a model instead of a correlation coefficient. The honest conclusion: Rating is most likely driven by factors this dataset doesn't capture at all actual app quality, user experience, crash rate, or competitive positioning not by installs, reviews, price, size, or pricing model.

Business Insights

Everything up to this point has been building toward two deliverables: Dashboard 1, the core analytics view covering catalogue composition, ratings, correlations, sentiment, and model results; and Dashboard 2, a dedicated view built around the six internship-assigned business tasks. Together, they translate this report's findings into something a stakeholder can explore directly, rather than read line by line.



Figure 9: Dashboard 1 (Full Overview)

Figure 9 shows Dashboard 1 in full, the delivery view for every insight discussed in this chapter.

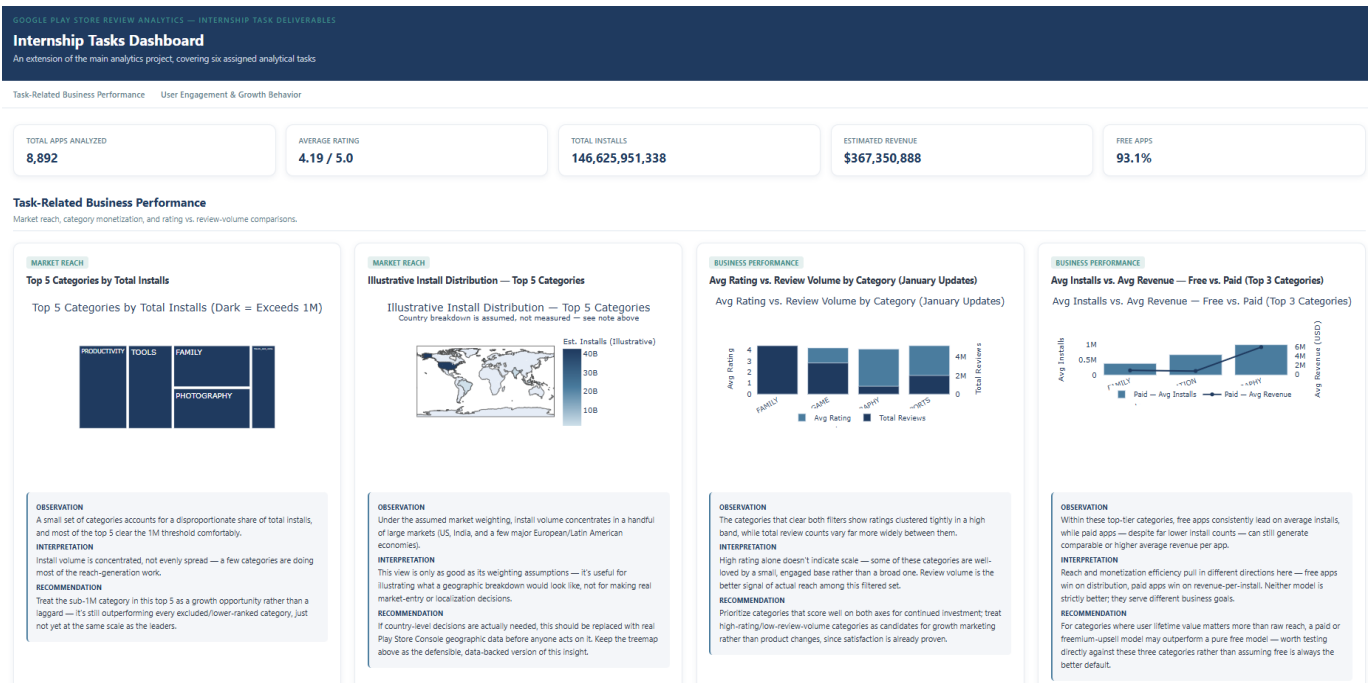


Figure 10a: Dashboard 2 (Internship Tasks Overview - KPIs & Business Performance)

Figure 10a shows Dashboard 2's KPI row and Task-Related Business Performance tab, covering Tasks 2, 5, and 6.



Figure 10b: Dashboard 2 (Internship Tasks Overview - User Engagement & Growth)

Figure 10b shows Dashboard 2's User Engagement & Growth Behavior tab, covering Tasks 1, 3, and 4.

Insight 1: Reach and Revenue Diverge

- Business Meaning: The category with the most installs isn't the category generating the most estimated revenue.
- Business Impact: Category investment decisions made on install volume alone risk missing where the actual revenue is.
- Decision Supported: Treat reach and monetization as two separate strategic tracks.
- Potential Action: Score category expansion proposals against both metrics independently.

Insight 2: Sentiment and Rating Don't Move Together

- Business Meaning: Review sentiment shows only a weak relationship with app ratings, indicating that written feedback and star ratings reflect different aspects of user experience.
- Business Impact: Ratings alone may overlook emerging customer concerns.
- Decision Supported: Track sentiment and ratings as separate performance indicators.
- Potential Action: Monitor review sentiment alongside ratings to identify issues earlier.

Insight 3: Paid Apps Rate Modestly Higher

- Business Meaning: Paid apps carry a slightly higher median rating (4.40) than free apps (4.30).
- Business Impact: The gap is real but too small to use as a standalone pricing-quality argument.
- Decision Supported: Pricing strategy shouldn't lean on this gap alone it's a minor signal, not a strong one.
- Potential Action: Pair this finding with category-specific revenue data (Insight 6) before drawing pricing conclusions.

Insight 4: Rating Resists Prediction

- Business Meaning: The machine learning model showed that ratings cannot be reliably predicted using installs, reviews, price, or size alone.
- Business Impact: App ratings are influenced by factors beyond the available metadata.
- Decision Supported: Avoid relying solely on marketplace metrics to improve ratings.
- Potential Action: Include richer product and user experience data in future predictive models.

Insight 5: Growth Comes in Bursts

- Business Meaning: Across the Task 3 and Task 4 time-series views, a small number of months carry a disproportionate share of each category's install growth.
- Business Impact: "Steady growth" is often a few strong months averaging out flatter ones, treating growth as continuous risks misreading what's actually driving it.
- Decision Supported: Growth should be investigated month-by-month, not assumed to be organic and constant.
- Potential Action: Cross-reference the highlighted spike months against release notes or marketing campaign dates to identify the actual driver.

Insight 6: Free Wins Reach, Paid Can Win Revenue-Per-App

- Business Meaning: Within top-performing categories, free apps lead decisively on installs, while paid apps can generate comparable or higher average revenue per app despite far lower install counts.
- Business Impact: Neither pricing model is universally superior, they serve different business goals.
- Decision Supported: The free-vs-paid choice should depend on whether the goal is reach or per-user monetization.
- Potential Action: For categories where lifetime value matters more than raw reach, test a paid or freemium-upsell model directly against a free-only baseline.

Recommendations

#	Recommendation	Source Insight
1	Evaluate category investment on reach and monetization separately	Insight 1
2	Don't use "reduce app size" as a default rating-improvement lever	Insight 3
3	Track sentiment and star rating as independent, parallel signals	Insight 2
4	Cross-reference high-growth months against release/campaign dates	Insight 5
5	Replace the illustrative Task 2 geography with real Play Console data before market decisions	—
6	Test a paid/freemium model directly against free-only in high-LTV categories	Insight 6

Act Now: supported directly by this analysis, no further data needed

- Evaluate category investment decisions on two separate tracks: reach (installs) and monetization (revenue), not one combined ranking.
- Stop treating app-size reduction as a default rating fix; the data shows no meaningful size-rating relationship among apps with proven install traction.
- Track review sentiment and star rating as two independent, parallel signals rather than one blended "satisfaction" score.

Investigate Further: directionally supported, but needs additional groundwork before acting

- Cross-reference the highlighted high-growth months (Tasks 3 and 4) against actual release notes or marketing campaign dates to confirm what drove each spike.
- Before making any real market-entry or localization call, replace the illustrative Task 2 geographic view with real, Play Console-sourced country-level install data.
- In categories where user lifetime value matters more than raw reach, run a direct test of a paid or freemium-upsell model against a free-only baseline rather than assuming free is the safer default.

Notes: No analysis is without limitations, and this one is no exception:

- Challenges faced: inconsistent raw data formats requiring custom parsing; no real geographic data for Task 2; heavily stacked multi-condition filters on Tasks 1 and 6; building reusable IST-based scheduling logic; and resisting the temptation to overstate a near-zero R^2 result.
- Key assumptions: missing Ratings were treated as non-recoverable rather than imputed; "Last Updated" was used as a proxy for install-growth timing (no true install-date field exists); Task 2's country weights are explicitly illustrative, not measured.

- Limitations: this is a static snapshot, not real-time data; Revenue is an estimate ($\text{Price} \times \text{Installs}$) that excludes in-app purchases and ads; VADER sentiment scoring is rule-based and won't catch sarcasm or mixed-tone reviews.

Future Scope

Extension	Resolves
Real, Play Console-sourced geographic install data	Task 2's illustrative country weights
Richer model features (description text, screenshots, update frequency, crash/ANR rates)	Rating's near-zero R^2 — genuinely new signal, not a bigger version of the same features
Live, refreshable dashboard on an updating data source	The dataset's static, single-snapshot nature
A/B testing specific recommendations (e.g., free vs. paid/freemium by category)	Recommendations currently based on observational correlation, not a controlled test
Deeper review-theme segmentation, beyond one compound sentiment score	VADER's inability to capture sarcasm or mixed-tone reviews

Short-Term

- Replace Task 2's illustrative geographic weighting with real country-level install data, once available.
- Segment review text further (e.g., clustering by theme) rather than relying on a single compound sentiment score per review.

Longer-Term

- Enrich the rating-prediction model with features this dataset simply doesn't have description text, screenshots, update frequency, crash/ANR rates since the current feature set has already been shown, twice over, not to explain rating variation.
- Move from a static, notebook-generated HTML dashboard to a live dashboard connected to a refreshing data source, deployed behind proper access controls.
- Run controlled A/B tests (e.g., paid vs. freemium in a specific high-LTV category) to validate recommendations that are currently supported observationally, not experimentally.

Conclusion

Question	Answer
Do reach and revenue lead to the same categories?	No — GAME leads installs, FAMILY leads revenue. Treat them as separate strategic tracks.
Does review sentiment agree with star rating?	Only weakly — confirmed independently through correlation analysis, sentiment scoring, and predictive modelling.
Can app rating be predicted from measurable attributes?	Not meaningfully ($R^2 \approx -0.036$) — rating is most likely driven by factors this dataset doesn't capture, such as actual product quality.

This project turned a raw, inconsistently formatted dataset into a tested, evidence-backed view of what actually drives success on the Google Play Store and, just as importantly, what doesn't. Every finding in this report survived at least one deliberate attempt to disprove it, whether through correlation, an independent sentiment signal, or a predictive model. That's the difference between a report that describes a dataset and one a stakeholder can actually build a decision on.